

Robust LP for optimal supply-chain management in biomethane plants

Project Proposal

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I. BACKGROUND

Biomethane production is going to cover the 4% of total energy consumption (TEC) of EU by 2050, due to the incentives of the EU New Green Deal that pushes the energy mix toward low-carbon, renewable and self-produced sources. In the next decade, anaerobic digestion biogas facilities usually equipped with combined heat&power engines for the main goal of electricity production, are going to be *upgraded* to biomethane production plants that can directly send their output to the national grid for third use. Since biomethane is easier to be stocked than electricity, the focus of plant managers is not going to track a market-driven or demand-driven production, but rather to maximize it. As a matter of fact, the incentive scheme (implemented in Italy by "DM biometano 15 settembre 2022") has promoted priority of dispatch and fixed tariffs until 2030 (110 €/MWh, almost double the average methane price in normal market conditions). Nevertheless, finding the best techno-economic (TE) trade-off point of operation is all but a trivial task. First of all, co-digestion of different organic substrates (es. energy crops, zootechnical wastes, organic fraction of urban waste) out-competes mono-digestion in terms of performances, but the degree of complexity in finding the best TE operating point increases substantially.

For a successful operation, the plant managers shall be also able to move within the recently-growing agro-industrial by-product (*upstream*) market, where co-substrates are purchased and which rules are all but well-established.

The anaerobic co-digestion (AcoD) process entails very non-linear and sometimes non-smooth kinetics; in addition, to have a comprehensive model able to grasp the behaviour of different complex (es. energy crops and slurries) co-substrate, the state-of-the-art modelling practice requires more than 40 state variables (IWA anaerobic digestion model n.1 (ADM1)). For this reason, the background idea was to decouple the upstream market economic optimization from the "in-plant" technical management optimization.

II. DETERMINISTIC LP DESCRIPTION

The facility is thus considered as an "intermediate conversion" layer between an *upstream* market of waste/byproducts (multiple inputs) and the *downstream*

market of energy retailing (single output). The plant is a priori authorized to transform only a subset of the available types of items available on the market. Decision variables are the amount of each item (*i-th* co-substrate) to be purchased from each supplier (*j-th* farmer, agro-industrial facility) to maximize the potential profit that may arise from an optimal management of the purchased quantities. In other words, the aim is to find the potentially optimal co-digestion diet, designed to be constant over the purchase horizon (*N* days of operation). No time domain is considered.

Objective function: maximisation of the profit from the theoretical amount of output i.e. "infinite" biomethane potential ($BMP_{\infty} (NL_{ch4} kg_{VS_i}^{-1})$, where VS stands for "volatile solids") made available at the plant, that can be produced with the stock of co-substrates purchased and to be fed to the reactor until the next purchase horizon. The $BMP_{\infty,i}$ is a "conversion yield" that can usually be found in literature and is obtained for every co-substrate by dedicated and standardized batch lab-tests: it is the asymptotic value of the cumulative amount of methane produced out of a certain amount of VS_i .

Constraints:

- *Stock capacity.* Upper bound on $\sum_{j=1}^m x_{i,j}$ can be potentially imposed. Also, if some stock is already/still present at the beginning of the purchase horizon, lower bounds can be enforced to guarantee its consumption in the next operation regime (null or even negative costs may be associated to those items j-indexed by "my plant").
- *Seller items availability.* The amount of $x_{i,j}$ that can be purchased by the *j-th* seller must be upper bounded by its availability.
- *Hydraulic retention time.* The "processable" total flowrate $\left(\frac{\sum_{i=1}^n \sum_{j=1}^m x_{i,j}}{N} \right) (m_{FM}^3 d^{-1})$ of the influent co-substrate mixture shall be limited with box constraints set by expert knowledge on process stability and legislative limitation imposed by the authority over the reactor waste discharge (i.e. amount of digestate produced, that has usually to be spread over a certain amount of available land hectares (legislative limit)).
- *Technical constraint.* A set of *K* technical limitations defined by heuristic knowledge shall be applied to find a co-digestion diet that shall be able to lead to stable operation. They are based on typically available substrate characterization and are applied in the form of box constraints over the weighted average of the *k-th* characteristic of the co-substrates mixture.

$$\begin{aligned}
\max_{x \in \mathcal{X}} \quad & \sum_{i=1}^n \sum_{j=1}^m [p_{ch4} BMP_{\infty,i} V S_i x_{i,j} - (C_{Pi,j} + C_{Ti} d_j) x_{i,j}] \\
\text{s.t.} \quad & \sum_{j=1}^m x_{i,j} \leq S_i \quad \forall i = 1, \dots, n, \\
& x_{i,j} \leq A_{i,j} \quad \forall i = 1, \dots, n, \quad \forall j = 1, \dots, m \\
& k_{min} \leq \frac{\sum_{i=1}^n \sum_{j=1}^m k_i x_{i,j}}{\sum_{i=1}^n \sum_{j=1}^m x_{i,j}} \leq k_{max} \quad \forall k = 1, \dots, K \\
& HRT_{min} \leq \frac{V_R N}{\sum_{i=1}^n \sum_{j=1}^m x_{i,j}} \leq HRT_{max}, \\
& x_{i,j} \geq 0 \quad \forall i = 1, \dots, n \quad \forall j = 1, \dots, m
\end{aligned}$$

where $x_{i,j}$ are the amount of purchased co-substrates (kg_{FM} where FM stands for "fresh matter"); $C_{Pi,j}$ and C_{Ti} are the specific purchase ($\text{€}kg_{FM_i}^{-1}$) and transportation costs ($\text{€}kg_{FM_i}^{-1}km^{-1}$) respectively; V_R is the reactor volume (m^3), N is the number of days in the purchase horizon (d), $A_{i,j}$ are the upper bounds for sellers' availabilities and S_i are the maximum in-plant stock capacities; p_{ch4} is the biomethane selling price ($\text{€}Nm_{ch4}^3$) and $V S_i$ ($kg_{VS_i}kg_{FM_i}^{-1}$) are the content of organic matter in the purchased item (the rest being water + inorganic matter). $HRT_{min}, HRT_{max}, k_{min}, k_{max}$ are scalar boundaries. k_i are the substrate characteristics subject to box constraints over the purchased mixture i.e. $k = \{\text{Total Kjeldahl Nitrogen (TKN), VS, lipid content, carbon/nitrogen content (C/N)}\}$. Unitary density for all items is considered.

III. UNCERTAINTY DESCRIPTION

- *Item purchasing cost from the seller.* If a contract exist between the supplier of organic substrate and the facility, it is unlikely it would enforce a constant purchasing cost. Contracting forms are most likely to fix lower and upper bounds. In other cases, the purchasing cost depends on the equilibrium of other markets that can be highly volatile.
- *Item availability of the seller.* Especially in the case of agro-industrial by-products, the amount of available item to be sold to the plant manager depends on the production of the supplier for its own primary market. If a form of contracting exists, it is most likely to fix lower and upper bounds for the amount that must be supplied.
- *$BMP_{\infty,i}$.* The values that can be found in literature are characterised by a certain variance because of a certain degree of variability in the testing conditions and the different specific substrates that are lumped under the i -th item umbrella (es. different cultivar of maize silage or maize silage growth in different times of the year are lumped within the "maize silage" i-indexing).